# DataSentinel — Pipeline Quality Report

Pipeline: **E-Commerce Orders Validation** | Target Source: Primary | Run ID: run-174478ad | Date: 2026-11-06 18:28:18 UTC

| Quality Pass Rate | Total Records | Failed Rules | Anomalies | Schema Drift |
|---|---|---|---|---|
| 100.0% | 30 | 0 | 0 | STABLE |

## 1. Quality Rules Validation Breakdown

| Status | Rule ID | Type | Column | Failures | Fail % | Details |
|---|---|---|---|---|---|---|
| PASSED | RULE-ORD-001 | null_check | order_id | 0/30 | 0.0% | Column 'order_id' has zero null values as required. |
| PASSED | RULE-ORD-002 | unique_check | order_id | 0/30 | 0.0% | Column 'order_id' uniqueness rate 100.00% meets requirement (100.00%). |
| PASSED | RULE-ORD-003 | null_check | customer_email | 0/30 | 0.0% | Column 'customer_email' has zero null values as required. |
| PASSED | RULE-ORD-004 | regex | customer_email | 0/30 | 0.0% | All non-null values in 'customer_email' match pattern '^[a-zA-Z0-9_.+-]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-.]+$&#... |
| PASSED | RULE-ORD-005 | range | total_amount | 0/30 | 0.0% | All values in 'total_amount' satisfy range requirement (>= 0.01 and <= 10000.0). |
| PASSED | RULE-ORD-006 | foreign_key | payment_method | 0/30 | 0.0% | All values in 'payment_method' exist in set of 4 reference values. |

## 2. Statistical Anomaly Detection (Z-Score & IQR)

| Column | Method | Outliers | Outlier Rate | Distribution Metrics | Top Anomalous Samples |
|---|---|---|---|---|---|
| **total_amount** | ZSCORE | 0 | 0.00% | μ=108.5047, σ=60.4409 | None |
| **total_amount** | IQR | 0 | 0.00% | Q1=65.75, Q3=143.75 | None |
| **item_count** | ZSCORE | 0 | 0.00% | μ=2.0333, σ=0.9279 | None |
| **item_count** | IQR | 0 | 0.00% | Q1=1.0, Q3=3.0 | None |